

Jason Zheng

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EDUCATION

Rensselaer Polytechnic Institute

Expected December 2027

B.S. Computer Science & Information Technologies and Web Science

GPA: 3.68/4.0

- Relevant Coursework: Computer Architecture & OS, Web Systems Development, Data Structures & Algorithms, Principles of Software

EXPERIENCE

AgileRadioCom — Software Development Intern

May 2026 – Present

Flask, Socket.IO, SSH/SFTP Automation, Real-Time Systems

- Built Python Flask web server with Socket.IO to orchestrate 5+ networked devices, managing complex workflows between Raspberry Pi, FPGA, and RTL-SDR hardware.s
- Implemented SSH/SCP/SFTP automation layer with robust error handling, status monitoring, and health checks for multi-device coordination
- Designed bidirectional real-time communication system managing state across distributed devices to prevent interference and ensure message ordering.

MercerX Lab, *BeyondGPA* — Backend Developer

Jan. 2026 – May 2026

Database Systems, NextAuth, NextJS Middleware, Prisma, Docker

- Prototyped an API rate limiting service using Next.js Middleware to prevent malicious usage and enforce strict quota enforcement
- Deployed production website to Rensselaer Polytechnic Institute's network, providing 400+ users with an organized platform to publish projects and discover research.
- Managed database schema and queries using Prisma ORM with PostgreSQL backend.

PROJECTS

SyllabusAI | *FastAPI, RAG (Groq + Gemini), Next.js, NextAuth, Docker*

Jan. 2026 – May 2026

- Architected a full-stack RAG application end-to-end, designing the MVP, decomposing features into sprint milestones, and owning delivery from prototype to deployment.
- Built a retrieval-augmented generation pipeline using Groq + Gemini APIs to parse and structure unformatted data into normalized, queryable tables via REST endpoints.
- Engineered a token-based rate limiting system with IP fallback to control API call volume, preventing abuse while maintaining UX for authenticated users.

Go Gaia | *Microsoft Azure Computer Vision, Flask, Git/GitHub Collaboration*

Sept. 2025

- Trained a **Microsoft Azure Custom Vision** model to classify the recyclability of an object through an image at **~60% accuracy**.
- Refactored the training process from **classification** to **detection** for better accuracy and leading to ~600 training cases.
- Utilized **Git** for version control and **GitHub** to effectively develop the application with the Team of 3 others.
- Received award from Avanade for best use of Microsoft Azure's AI.

Data Pulse | *Flask, Pandas, React, Bloomberg Terminal*

Sept. 2025 - Dec. 2025

- Designed and implemented a data routing pipeline to parse 10M+ rows of OHLCV stock data across 4,000+ equities, ingesting a decade of historical records from Bloomberg Terminal.
- Reduced average data lookup and filtration time from 20 minutes to 5 minutes (75% improvement)
- Built a full-stack dashboard with React + Recharts enabling both tabular and graphical visualization of filtered datasets via a Flask REST API backend.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS, R, MATLAB

Frameworks & Libraries: React, Node.js, Flask, JUnit, jQuery, FastAPI, Tailwind, PyArrow, Stable-Baselines3

Tools: Git, Docker, VS Code, Visual Studio, GenAI, Postman API, Bloomberg Terminal